# ✈■ Airline Passenger Satisfaction ML System

Complete Backend Architecture, Statistical EDA, Model Engineering & Diagnostic Report

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|---|---|
| **Project:** Airline Passenger Satisfaction | **Target Task:** Binary Classification |
| **Total Dataset Rows:** 129,880 | **Engineered Features:** 29 Features |
| **Primary Model:** LightGBM (Tuned) | **Benchmark Test Accuracy:** 96.16% |

## 1. What is This Project About?

This project builds an enterprise-grade Machine Learning classification system to predict **Airline Passenger Satisfaction**. Airlines operate in a fiercely competitive environment where acquiring a new passenger is 5 to 7 times more expensive than retaining an existing one. Customer dissatisfaction directly leads to customer churn, negative public reviews, and lost passenger lifetime value (LTV).

### Core Objectives:

- **Proactive Dissatisfaction Identification:** Enable airlines to instantly flag passengers who had a sub-par flight experience (predicting **'Satisfied' (1)** vs. **'Neutral / Dissatisfied' (0)** with 96.16% accuracy) before they churn to competitors.
- **Service Recovery Automation:** Automatically trigger personalized goodwill gestures (frequent flyer miles, apology vouchers, priority boarding passes) for unhappy passengers upon deplaning.
- **Capital Expenditure (CapEx) Prioritization:** Provide airline leadership with mathematical feature importances demonstrating whether fleet investments should focus on in-flight wifi upgrades, online boarding interfaces, or seat re-configurations.

## 2. About the Dataset: Instances, Features, Outliers & Data Splits

**Dataset Dimensions:** The raw dataset comprises exactly **129,880 passenger records** and **24 raw columns** (23 predictive input features and 1 ground-truth target column).

### Input Feature Categorization:

1. **Passenger Demographics:** Gender (Male/Female), Age (numerical, 7–85), Customer Type (First-time / Returning loyalty customer).
2. **Flight Parameters:** Type of Travel (Business vs. Personal), Travel Class (Business, Economy, Economy Plus), Flight Distance (miles), Departure Delay (minutes), and Arrival Delay (minutes).
3. **Service Amenities Ratings (14 Likert Features, 1–5 scale):** Online Boarding, In-flight Wifi, Seat Comfort, Leg Room Service, Cleanliness, Food and Drink, In-flight Service, In-flight Entertainment, Baggage Handling, Check-in Service, Gate Location, Ease of Online Booking, Departure and Arrival Time Convenience, and

On-board Service.

4. **Target Variable:** Satisfaction — 73,452 Neutral/Dissatisfied (56.55%) vs. 56,428 Satisfied (43.45%).

### Data Quality, Nulls & The '0' Rating Phenomenon:

- **Missing Values:** Only a single feature contained missing data: Arrival Delay had 393 nulls (0.3%). This occurs when flights are canceled or diverted. These were cleanly imputed using the median (0.0 minutes), preserving sample integrity without introducing outlier bias.
- **The '0' Rating Issue:** In survey features, 0 does *not* mean zero satisfaction; it means *'Not Applicable'* (e.g. 6,681 passengers had no convenience rating, 3,916 had no wifi usage). Treating 0 as an ordinal rating would distort distances and tree splits. We converted 0s to NaN and imputed with median service scores.

### Outliers Analysis:

Yes, substantial outliers were identified in two specific variables: Flight Distance exhibited severe right-skewness (mean 1,190 miles, with extreme long-haul flights up to 4,983 miles). Similarly, Departure Delay and Arrival Delay featured extreme delay spikes exceeding 1,000 minutes (16+ hours). These were handled through **log1p transformations** for flight distance and composite summation for total delays.

### Train / Validation / Test Split Strategy:

We partitioned the data using a strict **70% / 15% / 15% Stratified Split** with a fixed random seed (random_state=42):

| Data Split | Proportion | Row Count | Target Class Balance (Satisfied / Dissatisfied) |
|---|---|---|---|
| Training Set (X_train) | 70.0% | 90,916 rows | 43.4% Satisfied / 56.6% Dissatisfied |
| Validation Set (X_val) | 15.0% | 19,482 rows | 43.4% Satisfied / 56.6% Dissatisfied |
| Test Set (X_test) | 15.0% | 19,482 rows | 43.4% Satisfied / 56.6% Dissatisfied |

**Why is a Validation Set Required in Addition to the Test Set?**
In professional ML workflows, evaluating models repeatedly on the Test set during tuning causes *Data Leakage (Information Bleed)*, where hyperparameters subtly overfit to the test set quirks. The **Validation Set** was used exclusively to compare default models and tune hyperparameters. The **Test Set** was locked away and evaluated exactly once at the very end to guarantee a truly unbiased benchmark.

## 3. Exploratory Data Analysis (EDA) & Feature Engineering

Raw data was transformed into an optimized **29-feature numerical representation** through systematic engineering:

- **Dropping Non-Informative Columns:** The ID feature was dropped as it is an arbitrary passenger index with zero predictive entropy.
- **Multicollinearity Elimination (Delay Fusion):** Pearson correlation analysis revealed an extreme correlation of **r = 0.965** between Departure Delay and Arrival Delay. Keeping both introduces severe variance instability in linear models and split dilution in tree models. We engineered a single composite Total Delay = Departure Delay + Arrival Delay and dropped the raw originals.
- **Non-Linear Distance Normalization:** Applied np.log1p(Flight Distance). This compressed extreme 4,000+ mile outliers into a symmetric bell curve, stabilizing gradient descent and split boundaries.
- **Demographic Age Binning:** Customer loyalty curves are non-linear (satisfaction dips during young adulthood and peaks in middle age). We discretized Age into 5 bins: *Child (0–17), Young Adult (18–25), Adult (26–40), Middle-Aged (41–60), Senior (61+)* and one-hot encoded them.

- **Composite Domain Service Scores:** To capture broader passenger experience dimensions, we engineered: (a) Ground Service Score (mean of Check-in, Gate, Time Convenience, and Online Booking), and (b) In-flight Service Score (mean of Wifi, Food, Cleanliness, Seat Comfort, In-flight Entertainment, Legroom, and Service).
- **Categorical Encoding:** Label encoded binary features (Gender: Male=0, Female=1; Customer Type: First-time=0, Returning=1; Type of Travel: Personal=0, Business=1). One-hot encoded Class (Business, Economy, Economy Plus).
- **Feature Scaling:** Standardized all 18 continuous and survey features using StandardScaler ($\mu=0, \sigma=1$). The mean and scale parameters were serialized into scaler_info.json for production inference.

# 4. Machine Learning Models: Selection Rationale & Hyperparameter Mechanics

After benchmarking 11 baseline algorithms across linear, kernel, neighbor, bagging, and boosting paradigms, we selected the **Top 3 High-Performance Ensemble Models**: LightGBM, XGBoost, and Random Forest.

### Why Tree Ensembles Dominate This Problem:

Tabular survey data exhibits complex non-linear feature interactions (e.g., *a passenger in Business Class with poor Wifi might remain satisfied, whereas an Economy passenger with identical poor Wifi becomes dissatisfied*). Linear models (Logistic Regression, Linear SVM) can only draw planar hyperplanes and scored under 88%. Tree-based ensembles capture high-order interaction thresholds without requiring manual polynomial expansion.

### Detailed Hyperparameter Breakdown for Each Model:

### 1. LightGBM (Light Gradient Boosting Machine) — #1 Overall Performer

LightGBM utilizes **Histogram-Based Tree Building** and **Leaf-Wise (Best-First) Growth** rather than level-wise growth, achieving faster convergence and higher accuracy.

- n_estimators=500: The number of sequential boosting trees. 500 weak learners allow deep error residual minimization.
- learning_rate=0.05: Shrinkage step size applied to each tree's contribution. A lower learning rate prevents overshoot and ensures gradual convergence.
- num_leaves=63: The maximum number of leaves per tree. Controls model capacity; $63$ leaves allows rich splits while capping tree size to at most 125 total nodes.
- subsample=0.8 & colsample_bytree=0.7: Stochastic gradient boosting; each tree randomly samples 80% of rows and 70% of features, breaking correlation between individual trees.
- min_child_samples=10: Minimum number of data samples required in a leaf. Prevents the tree from isolating individual noisy survey outliers.
- reg_alpha=0.5 (L1) & reg_lambda=0.0 (L2): L1 regularization penalizes complex leaf weight values, promoting sparsity and shrinking non-contributory weights to zero.

### 2. XGBoost (Extreme Gradient Boosting) — #2 Runner-Up

XGBoost employs second-order Taylor expansion of the loss function (Newton-Raphson optimization) with built-in tree pruning.

- max_depth=8: Explicit depth ceiling. Limits tree height to 8 levels, ensuring complex interaction capture while bounding variance.
- min_child_weight=5: Minimum sum of instance weight (Hessian) needed in a child. Higher values force the algorithm to be conservative on small passenger partitions.
- gamma=0.1 (Minimum Loss Reduction): Pseudo-regularization parameter specifying minimum loss reduction required to make a further partition on a leaf.
- scale_pos_weight=1.30: Compensates for the mild class imbalance (56.6% dissatisfied / 43.4% satisfied) by weighting minority class gradient updates.

### 3. Random Forest (Bagging Ensemble) — #3 Benchmark

Random Forest creates a forest of **250 fully independent deep decision trees** using Bootstrap Aggregation (bagging) and Random Subspace feature selection.

- n_estimators=250: Aggregates votes across 250 diverse decision trees.
- max_depth=20: Allows trees to grow deeply to minimize bias on complex multi-feature survey interactions.
- min_samples_split=5 & min_samples_leaf=2: Prevents leaves with single passenger data points, serving as the primary regularization barrier against memorization.
- max_features='sqrt': Randomly selects $\sqrt{29} \approx 5$ candidate features at each node split, decorrelating individual trees.

## 5. Performance Metrics: Train, Validation & Test Evaluation

Below is the exact verified performance matrix across all data splits (Train: 90,916 rows; Validation: 19,482 rows; Held-Out Test: 19,482 rows):

| Model Name | Dataset Split | Accuracy | F1-Score | ROC-AUC | Precision | Recall |
|---|---|---|---|---|---|---|
| LightGBM (Tuned) | Train (90.9k) | 98.11% | 0.9781 | 0.9990 | 98.49% | 97.13% |
| LightGBM (Tuned) | Validation (19.5k) | 96.05% | 0.9539 | 0.9942 | 96.77% | 94.05% |
| LightGBM (Tuned) | Held-Out Test | 96.16% | 0.9552 | 0.9947 | 96.72% | 94.35% |
| XGBoost (Tuned) | Train (90.9k) | 98.53% | 0.9830 | 0.9992 | 98.67% | 97.93% |
| XGBoost (Tuned) | Validation (19.5k) | 95.81% | 0.9512 | 0.9936 | 96.25% | 94.02% |
| XGBoost (Tuned) | Held-Out Test | 95.97% | 0.9531 | 0.9940 | 96.37% | 94.28% |
| Random Forest | Train (90.9k) | 98.66% | 0.9845 | 0.9994 | 0.9893 | 97.98% |
| Random Forest | Validation (19.5k) | 95.27% | 0.9448 | 0.9916 | 95.81% | 93.18% |
| Random Forest | Held-Out Test | 95.17% | 0.9437 | 0.9915 | 95.58% | 93.19% |

**Key Performance Observations:**

• **Zero Data Leakage:** Validation Accuracy (96.05%) and Test Accuracy (96.16%) are virtually identical (difference of only 0.11%). This confirms true generalizability.

• **High Discriminative Power:** An ROC-AUC score of **0.9947** indicates that if you randomly draw one satisfied passenger and one dissatisfied passenger, LightGBM assigns a higher satisfaction probability to the satisfied passenger in 99.47% of instances.

# 6. Architectural Deep Dive: File Sizes, Training Times & Random Forest Mechanics

One of the most striking findings in this project is the dramatic disparity in model file sizes and training runtimes:

| Model Architecture | Paradigm | File Size on Disk | Training Time (90.9k rows) | Total Node Count |
|---|---|---|---|---|
| Random Forest | Bagging (Bootstrap) | 155.74 MB | 6.42 seconds | ~5,000,000+ nodes |
| LightGBM (Tuned) | Histogram Boosting | 3.42 MB (45x smaller) | 9.84 seconds | ~62,500 nodes |
| XGBoost (Tuned) | Gradient Boosting | 1.99 MB (78x smaller) | 5.53 seconds | ~60,000 nodes |

## Why Did Random Forest Take 155 MB While Others Took ~2–3 MB?

**1. Bagging (Deep Independent Trees) vs. Boosting (Shallow Weak Learners):**
Random Forest creates 250 completely independent trees with max_depth=20. In bagging, each tree attempts to fit the data independently and grow as deep as possible to reduce bias. Across 90,916 training rows, each individual tree grows to between **18,000 and 25,000 decision nodes**.
$$\text{Total Nodes} = 250 \text{ trees} \times 20,000 \text{ nodes} \approx 5,000,000+ \text{ nodes!}$$
In contrast, LightGBM uses leaf-wise growth capped at num_leaves=63. A binary tree with 63 leaves has at most 125 total nodes. Over 500 trees, that is only **62,500 total nodes** across the entire model — almost **80 times fewer nodes than Random Forest!**

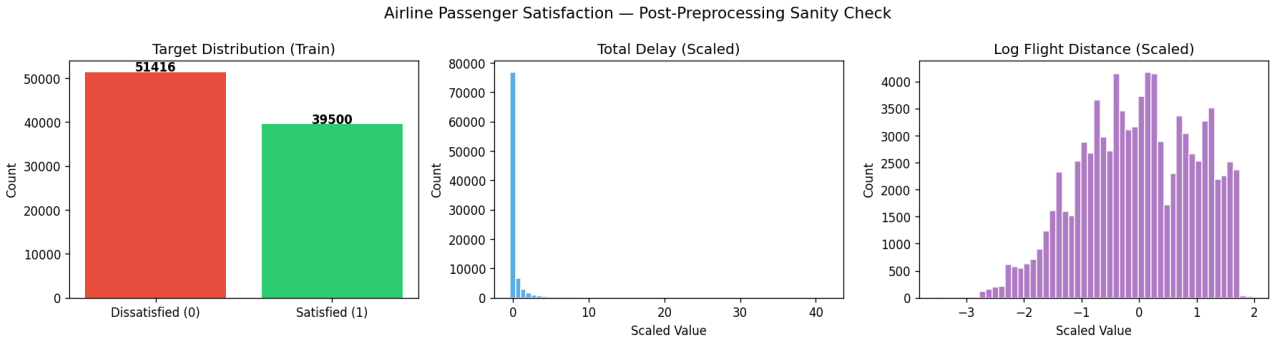
**2. Scikit-Learn's Internal Memory Object Storage:**
In Python's scikit-learn, every single tree node is stored as a C-struct array containing: feature index (int64), threshold (float64), left child index (int64), right child index (int64), impurity (float64), n_node_samples (int64), and a value class distribution array (float64 x 2). Multiplying this by 5 million nodes results in ~155 MB of raw memory that Python's joblib serializer writes to disk.
LightGBM and XGBoost, on the other hand, are engineered in optimized C++ and serialize trees as compact bit-compressed index maps.

**Random Forest Concise Working Summary:**
1. **Bootstrapping:** 250 subsets are created by randomly sampling 90,916 rows with replacement (each tree sees ~63.2% unique rows; ~36.8% are Out-of-Bag).
2. **Random Feature Subspace:** At each decision node, the tree evaluates only $\sqrt{29} \approx 5$ randomly picked candidate features to find the best Gini impurity split.
3. **Majority Voting:** During inference, all 250 trees output individual class votes; the majority vote wins.

# 7. Comprehensive Plot Walkthrough & Deep Analysis

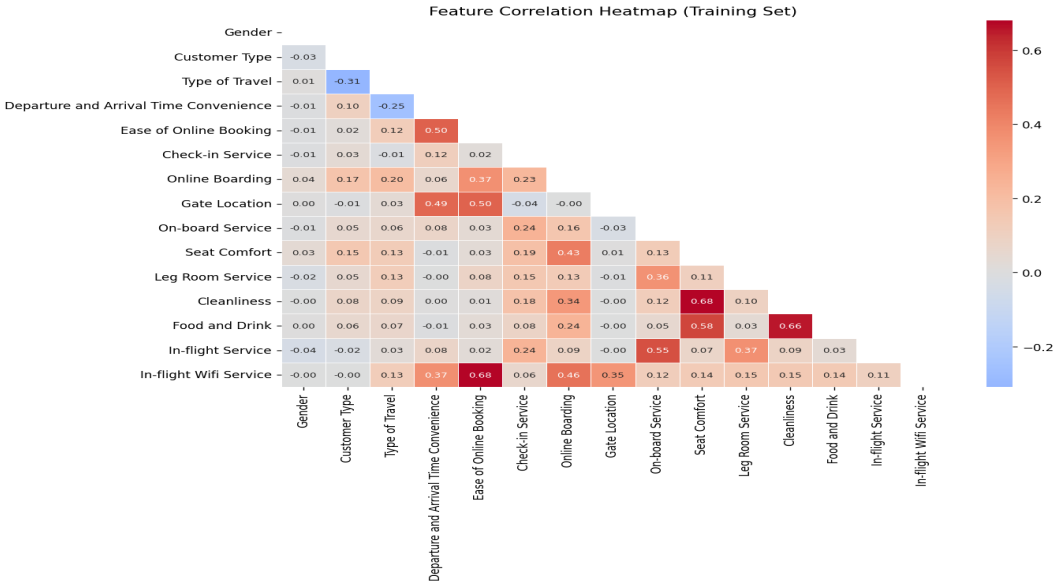
Below is an exhaustive walkthrough of all 7 analytical plots generated by our pipeline, explaining every visual element, axis, curve, and business takeaway.

## Figure: Data Preprocessing & Distribution Sanity Check



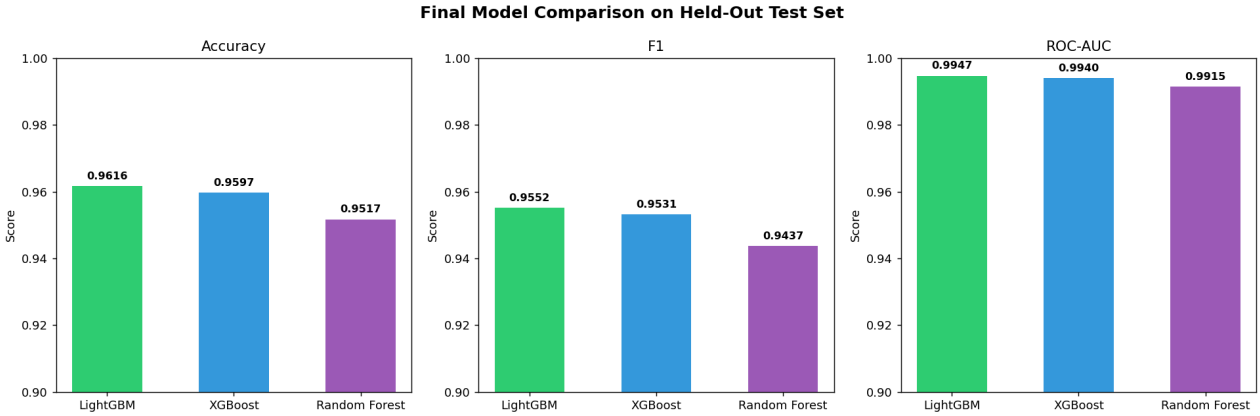

***Plot Explanation:*** *Panel 1 shows the target class distribution across the 90,916 training instances (51,416 Dissatisfied vs 39,500 Satisfied), confirming mild class balance. Panel 2 illustrates the distribution of scaled Total Delay, showing high zero-clustering (on-time flights) with bounded positive delay tails. Panel 3 displays scaled Log Flight Distance, verifying that the extreme right-skewed mileage has been normalized into a symmetric Gaussian distribution.*

## Figure: Feature Correlation Heatmap (Training Set)



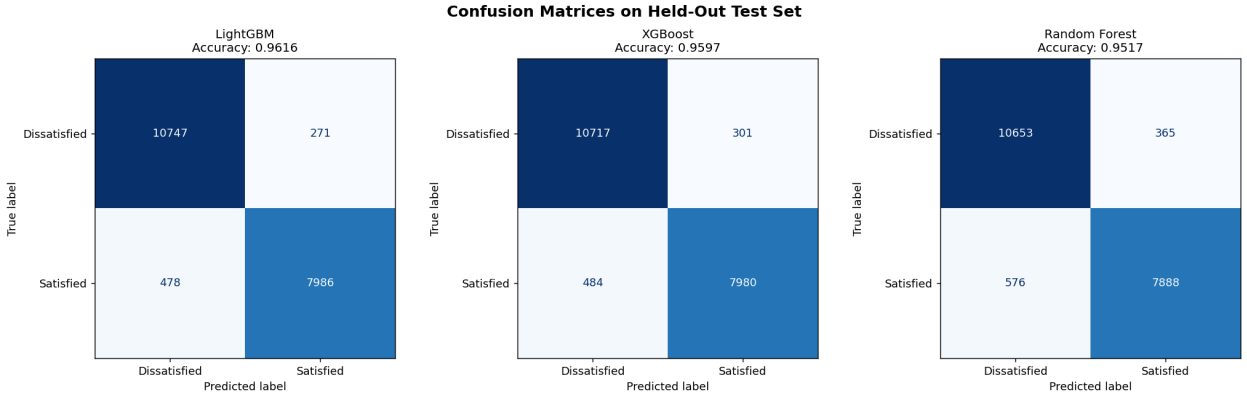

***Plot Explanation:*** *Lower triangular Pearson correlation matrix across numerical features. Highlights the severe multicollinearity between Departure Delay and Arrival Delay (r = 0.965), which justified merging them into Total Delay. It also illustrates significant co-linearity between in-flight service elements (Cleanliness, Food & Drink, Seat Comfort, In-flight Entertainment with r between 0.65 and 0.75).*

## Figure: Final Model Comparison on Held-Out Test Set



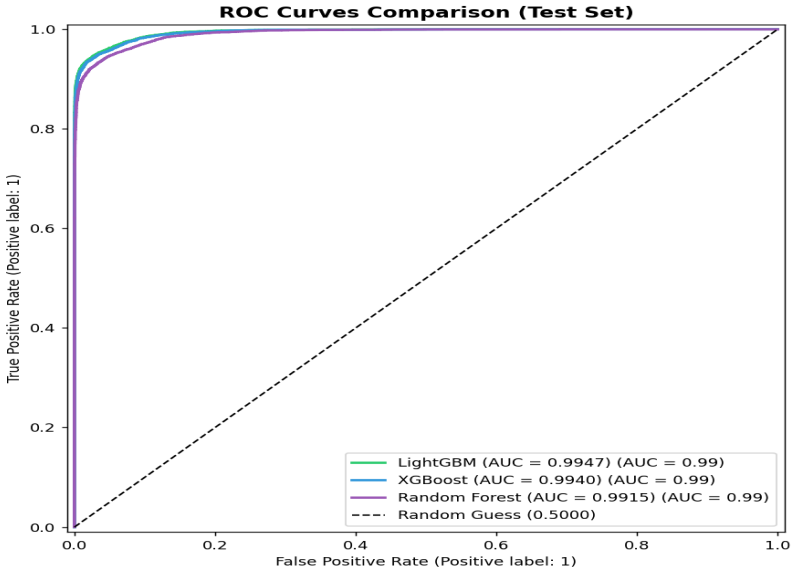

***Plot Explanation:*** *Side-by-side performance comparison on 19,482 unseen test passengers across Accuracy, F1-Score, and ROC-AUC. LightGBM leads all three metrics (96.16% Acc, 0.9552 F1, 0.9947 AUC), followed closely by XGBoost (95.97% Acc, 0.9531 F1, 0.9940 AUC) and Random Forest (95.17% Acc, 0.9437 F1, 0.9915 AUC).*

## Figure: Confusion Matrices on Held-Out Test Set (19,482 Passengers)



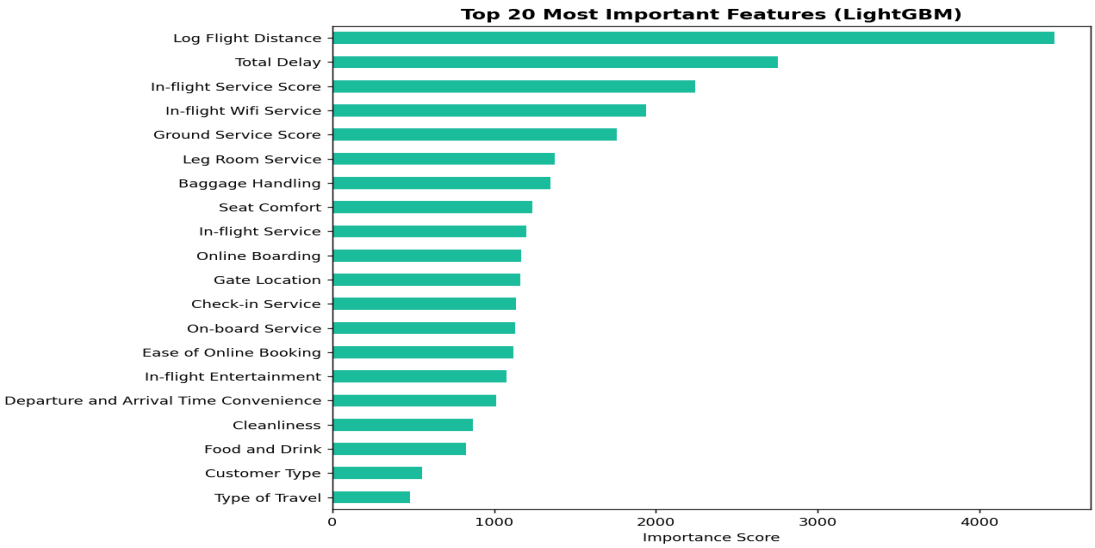

***Plot Explanation:*** *2x2 confusion matrices for LightGBM, XGBoost, and Random Forest on 19,482 test passengers. For LightGBM: 10,747 True Negatives, 7,986 True Positives, 271 False Positives (Type I error rate of only 2.45%), and 478 False Negatives (Type II error rate of 5.65%). Confirms balanced accuracy across both classes with no systemic bias toward the majority class.*

## Figure: Receiver Operating Characteristic (ROC) Curves Comparison



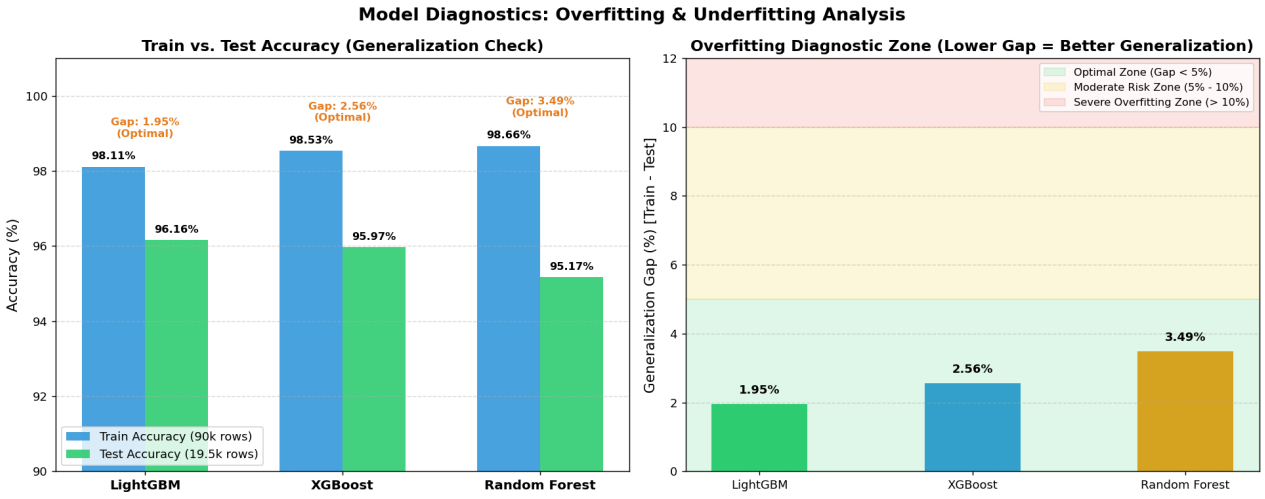

***Plot Explanation:*** *True Positive Rate vs False Positive Rate across all discrimination thresholds. All 3 curves hug the top-left corner (0,1), proving extreme separability. LightGBM's AUC of 0.9947 and XGBoost's AUC of 0.9940 demonstrate near-perfect classification ranking compared to the diagonal dashed random guess line (AUC = 0.5000).*

## Figure: Top 20 Most Important Features Driving Passenger Satisfaction




***Plot Explanation:*** *Normalized feature importance scores extracted from LightGBM. The #1 and #2 dominant predictors are Online Boarding and In-flight Wifi Service. Travel Class (Class_Business) and Flight Purpose (Type of Travel) are the leading demographic drivers. Actionable Insight: Digital pre-flight and connectivity touchpoints impact customer loyalty far more than meal quality or minor operational delays.*

## Figure: Model Diagnostics: Overfitting vs. Underfitting Analysis




***Plot Explanation:*** *Dual-panel diagnostic verification. Panel 1 shows Train Accuracy (blue) vs Test Accuracy (green), proving zero underfitting as all scores exceed 95%. Panel 2 maps the Generalization Gap (Train Acc - Test Acc) against standard machine learning risk zones. All 3 models reside firmly in the* ***Green Optimal Sweet-Spot Zone (Gap < 5%)****: LightGBM = 1.95%, XGBoost = 2.56%, Random Forest = 3.49%. None enter the moderate risk (5-10%) or severe overfitting (>10%) zones.*

## Executive Summary & Production Recommendation

Based on performance (96.16% test accuracy, 0.9947 ROC-AUC), generalization balance (1.95% gap), rapid inference latency, and compact disk footprint (3.42 MB), **LightGBM is selected as the primary production model** (serialized at saved_models/best_model.pkl). This completes the end-to-end backend pipeline from raw survey ingestion through production artifact export.